RIVER VALLEY HIGH SCHOOL

JC 2 PRELIMINARY EXAMINATION

CANDIDATE NAME	

CENTRE NUMBER	S					CLASS	21J	INDEX NUMBER					

BIOLOGY **9744/03**

Paper 3 Long Structured and Free-response Questions **13 September 2022**

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE ON ANY BARCODES.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
Total	

Section A

Answer **all** the questions in this section.

1 In plants, photosynthesis takes place in chloroplasts, which contain different types of photosynthetic pigments.

(a) Describe the role of different types of photosynthetic pigments in the photoactivation of chlorophyll. [3]

Scientists suggested that chloroplasts may have originated from prokaryotic cells that continue to function after becoming engulfed by primitive eukaryotic cells.

(b) Other than genomic features, describe **two** evidence that support this hypothesis. [2]

Saline soils, with high concentration of sodium ions (Na^+), are a major problem in many parts of the world. Most plant crop species are unable to tolerate high concentrations of sodium ions.

An experiment was conducted to determine the effect of salt (NaCl) concentration on the overall light-dependent reactions and separately on photophosphorylation activities involving photosystem I and photosystem II.

Table 1.1 shows the results of this experiment on a plant crop species.

Table 1.1

concentration of NaCl / mol dm^{-3}	photosynthetic activity / % activity relative to activity at 0 mol dm^{-3} of NaCl		
	photosystem I	photosystem II	overall light-dependent reactions
0.0	100	100	100
0.1	100	58	80
0.2	98	51	59
0.3	101	35	57
0.4	99	32	39
0.5	100	17	35

Adapted from El-Sheekh MM, Journal of Plant Physiology, 2004

(c) With reference to activities involving the photosystems, predict and explain what will happen to the growth of the plant as salt concentration continues to increase above 0.5 mol dm^{-3}. [3]

Recently one group of Na^+ transporters, HKT1, in root cells was found to be important for salt tolerance in plants. HKT1 actively removes Na^+ from the xylem sap. This prevents transport to and accumulation of Na^+ in the shoot.

Scientists have successfully isolated *HKT1* gene from salt-tolerant plants to create a *HKT1* gene construct as shown in Fig. 1.1.

pro	*HKT1*	ter

key

pro – root cell-specific promoter

ter – termination sequence

HKT1 – gene from salt-tolerant plant

Fig. 1.1

HKT1 gene construct was then used to produce salt-tolerant transgenic plant. A transgenic plant refers to a plant whose DNA is modified by inserting a gene from another source.

Fig. 1.2 shows the process of how salt-tolerant transgenic plants is formed.

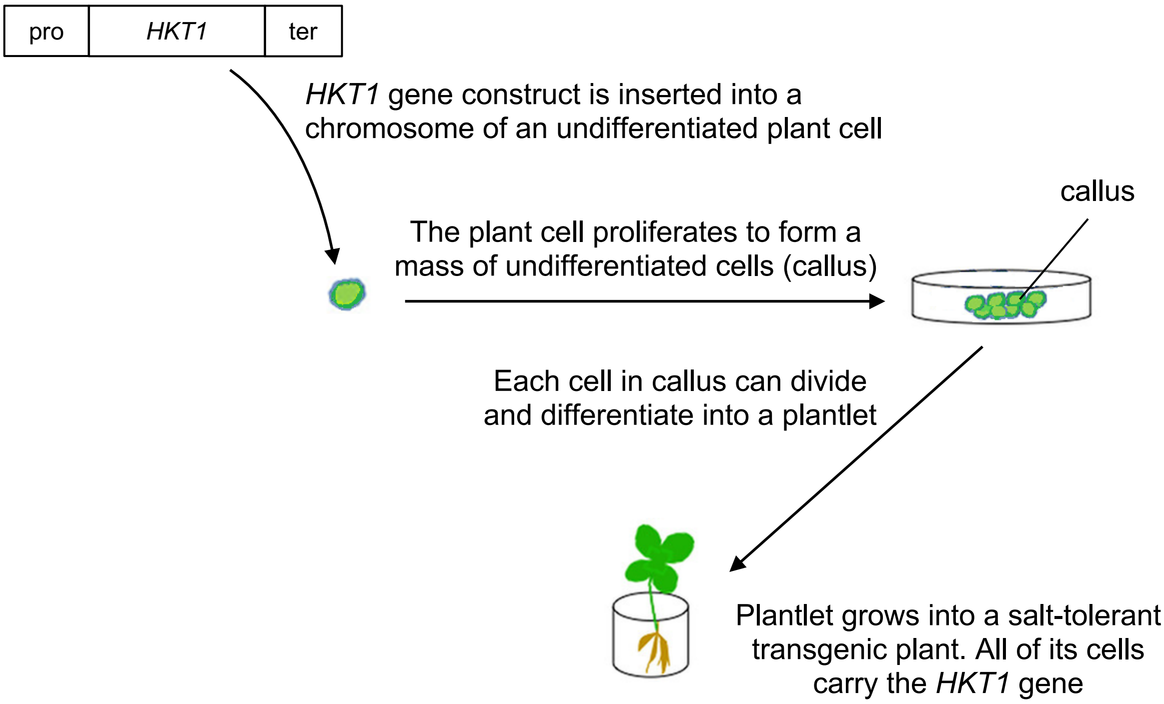

Fig. 1.2

(d) **(i)** Explain why the root cell-specific promoter was included in the *HKT1* gene construct. [2]

(ii) Cells from the callus in Fig. 1.2 are similar to zygotic stem cells in humans.

Explain why these callus cells can be used to produce the entire plant. [2]

Screening was conducted to verify that the 1.7 kb *HKT1* gene has been inserted into the transgenic plant's chromosome. The screening consisted of the following stages:

- DNA was isolated from the transgenic plant.
- The DNA was cut with restriction enzyme and the fragments produced were separated by gel electrophoresis.
- Detection of *HKT1* gene was then carried out.
- Fig. 1.3 shows the region where the radioactive probe binds to on the plant chromosome. The location of restriction site is indicated by *.

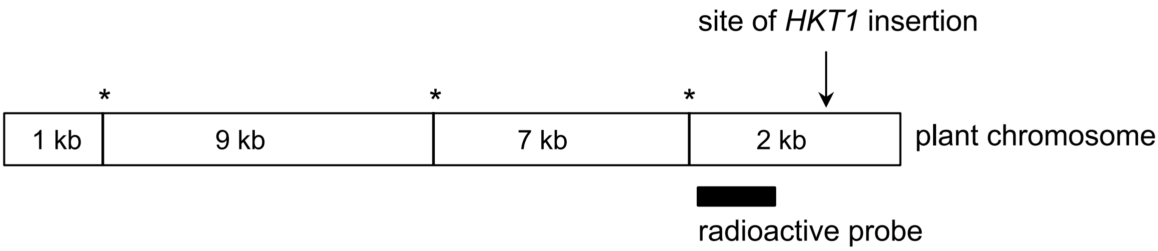

Fig. 1.3

(e) Outline how detection of *HKT1* gene was carried out. [3]

An experiment was performed on three groups of transgenic plants (T1, T2 and C) to investigate the effect of *HKT1* gene on plant growth in soils of different salinity. The three groups of plants were as follows:

T1	Transgenic plants containing one copy of the *HKT1* gene
T2	Transgenic plants containing two copies of the *HKT1* gene
C	Control plants without *HKT1* gene

The fresh weight of the plants was determined one week after growing in soils containing different concentrations of NaCl and is shown in Fig.1.4.

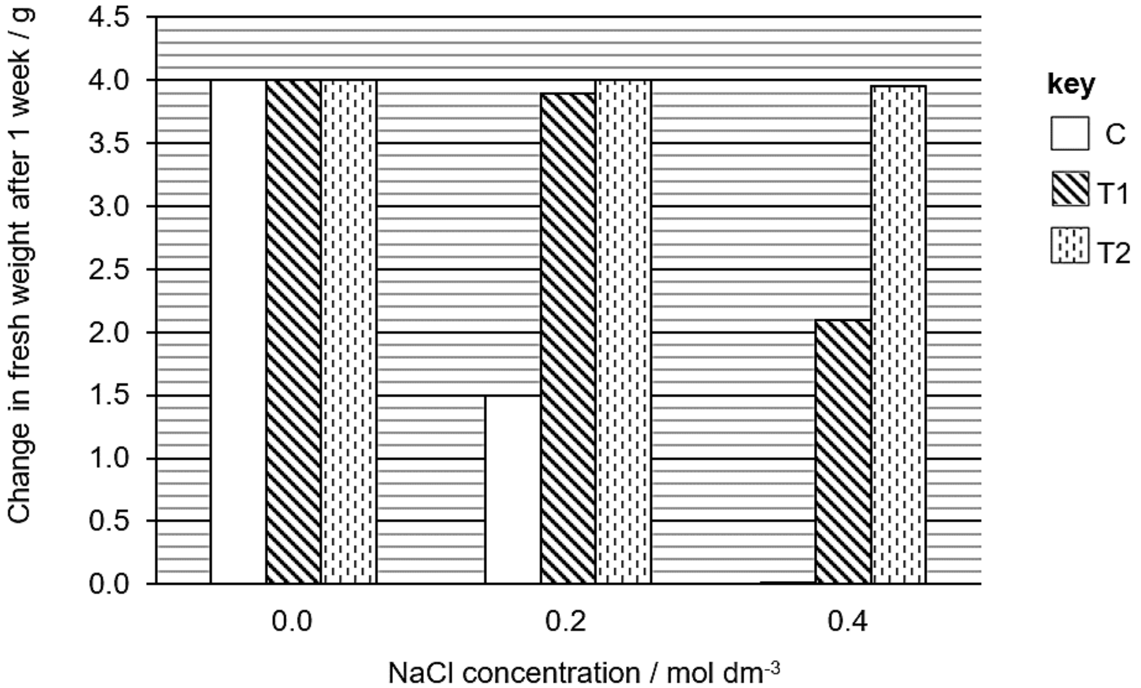

Fig. 1.4

(f) **(i)** With reference to Fig. 1.4, describe the effect of *HKT1* gene on the change in fresh weight of plants at each concentration of NaCl. [3]

(ii) explain the difference in the change in fresh weight of T1 and T2 plants. [1]

Fig. 1.5 is a schematic representation of the regions present in the HKT1 transporter, a membrane bound protein.

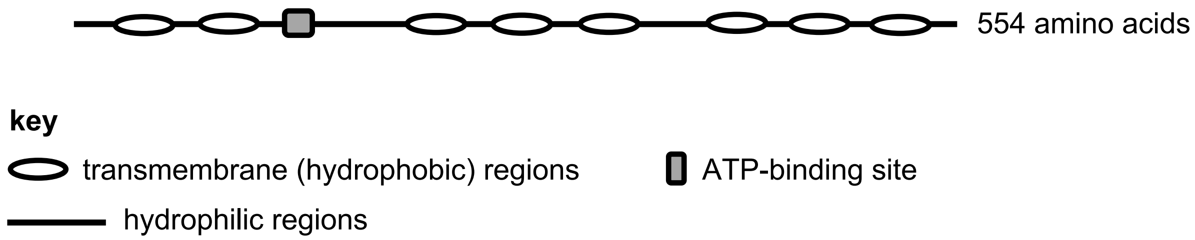

Fig. 1.5

(g) Explain how the different regions shown in Fig. 1.5 allow HKT1 transporter to perform its function. [3]

Certain plants can adapt to changes in environment by changing the amount of plant hormones, auxin and cytokinin, produced.

Fig. 1.6 shows the effect of drought and mineral nutrient deficiency on auxin and cytokinin biosynthesis in these plants.

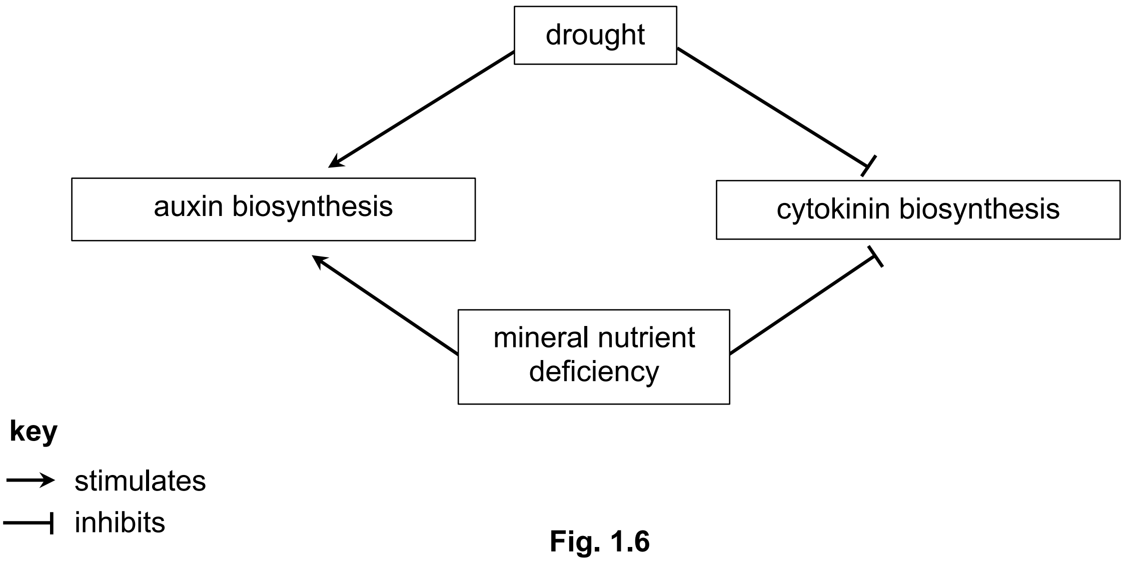

Fig. 1.6

The effects of various ratios of auxin to cytokinin on plant growth are summarised in Table 1.2. The extent of growth is shown on a scale increasing from + to +++.

Table 1.2

ratio of auxin to cytokinin	growth of shoot	growth of root
10:0	no growth	+++
8:2	no growth	++
6:4	no growth	+
5:5	no growth	no growth
4:6	+	no growth
2:8	++	no growth
0:10	+++	no growth

(h) Using Table 1.2 and Fig. 1.6, explain how these plants can survive under conditions of drought and mineral nutrient deficiency. [3]

[Total: 25]

2 An open reading frame (ORF) is the DNA sequence between a start and stop codon. ORFs are commonly used to predict the presence of genes.

Fig. 2.1 shows the same sample sequence and three possible ORFs due to different possible reading frames. Start codons are underlined and bolded, and stop codons are underlined and italicised.

1 **<u>ATG</u>** CAA TGG GGA AAT GTT ACC CTT ATT GAG GTA AGA CAG ATT *<u>TAA</u>*

2 A TGC ATT GGG GAA **<u>ATG</u>** TTA CCC TTA TTG AGG *<u>TAA</u>* GAC AGA TTT AA

3 AT GCA **<u>ATG</u>** GGG AAA TGT TAC CCT TAT *<u>TGA</u>* GGT AAG ACA GAT TTA A

Fig. 2.1

(a) Using your knowledge of genomic organisation, explain why the use of ORF is more effective in predicting genes in prokaryotic genome than in eukaryotic genome. [1]

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(b) Outline how the ORF region of sequence **2** may be amplified from a colony of bacteria cells. [4]

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Question 2 continues on page 12

Researchers investigated the possible relationships of ORFs and non-coding sequences with the prokaryote genome size.

Fig. 2.2 shows the relationship between **number of ORFs** and genome size.

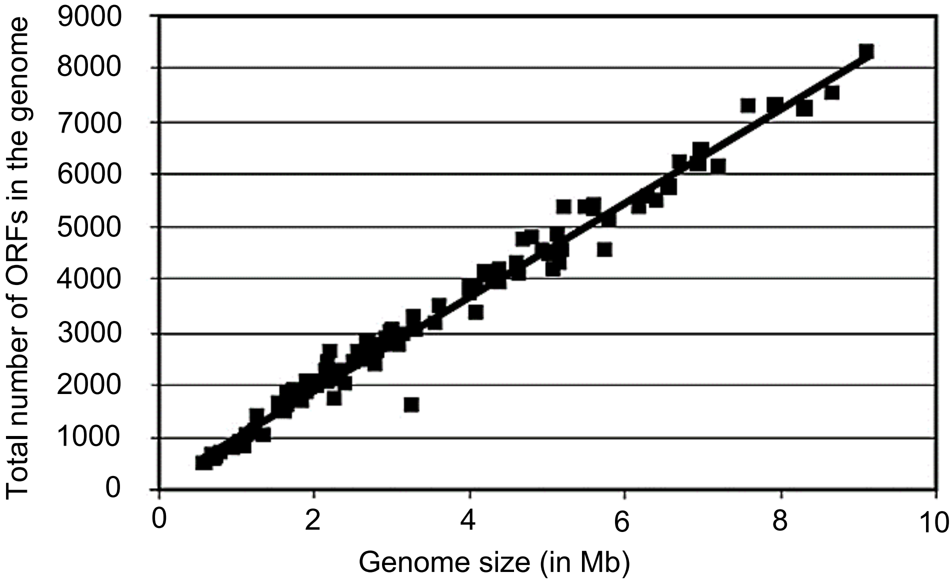

Fig. 2.2

Fig. 2.3 shows the relationship between the **size of non-coding DNA** and genome size.

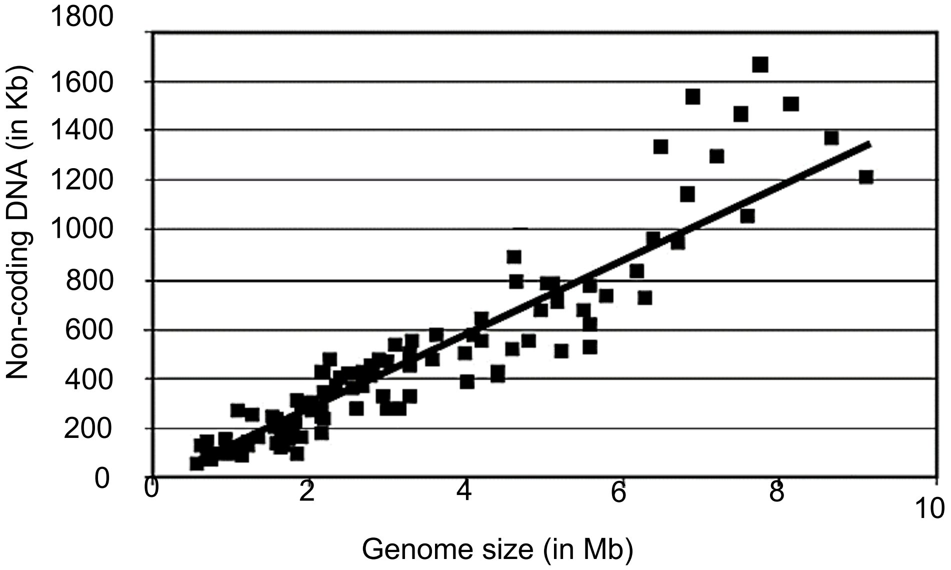

Fig. 2.3

Adapted from Konstantinidis and Tiedje, PNAS, 2004

(c) **(i)** Put a tick (✓) in one box to indicate which relationship shows a weaker correlation between the two variables investigated. [1]

Fig. 2.2 ☐ Fig. 2.3 ☐

(ii) With reference to your answer in **(c)(i)**, evaluate the extent to which the variable investigated can predict the genome size of the prokaryote. [3]

Obligate endocellular parasites are one of the smallest genome-sized prokaryotic species.

Fig. 2.4 shows how symbiotic relationship with their host affected genome size in terms of the number of genes and non-coding sequences.

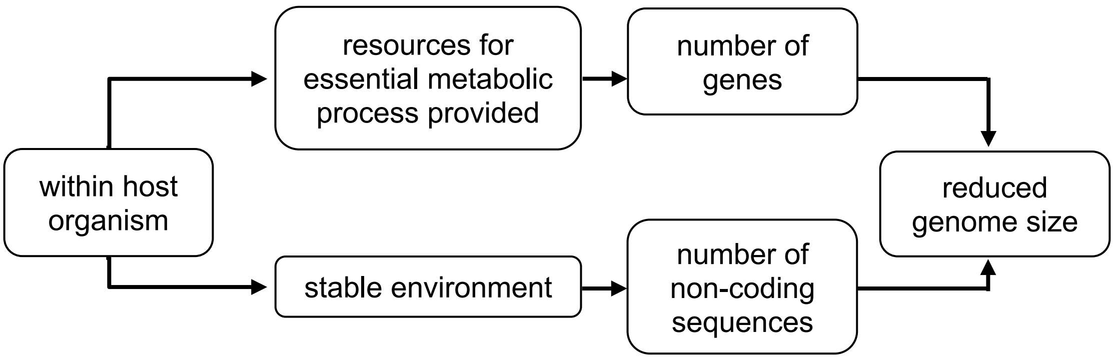

Fig. 2.4

(d) With reference to Fig. 2.4, explain how symbiotic relationship of obligate endocellular prokaryotes with host organisms results in a reduced genome size. [3]

Compared with RNA viruses, prokaryotes have a lower mutation rate.

(e) **(i)** Name an RNA virus. [1]

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(ii) Explain why prokaryotes have a lower mutation rate compared to RNA viruses. [2]

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[Total: 15]

3 Rattlesnakes contain a tailshaker muscle which causes a rattling sound upon contraction.

(a) Explain how oxygen uptake by muscle cells changes when the tailshaker muscle contracts. [3]

An investigation was conducted to study the tailshaker muscle and a mammalian muscle.

Fig. 3.1 shows the rate and sources of ATP supply in the tailshaker muscle and the mammalian muscle during contraction.

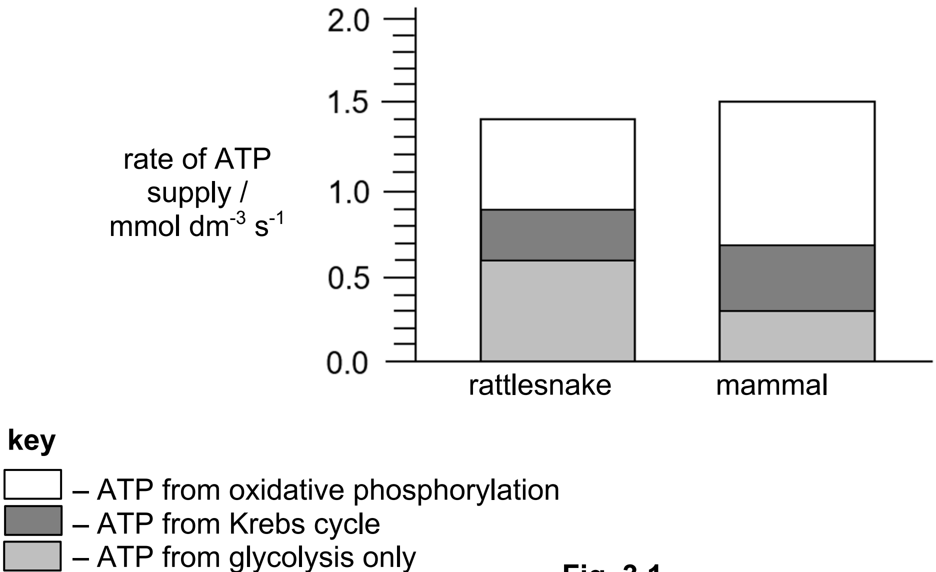

Fig. 3.1

(b) Using Fig. 3.1, calculate the amount of ATP supplied by anaerobic respiration in the tailshaker muscle in 1 minute.

Show your working clearly. [2]

Table 3.1 shows the changes in activities in the two muscles during contraction, relative to when the muscles are at rest.

Table 3.1

	change in blood flow / ml min^{-1}	change in O_2 uptake by muscle cells / $\mu mol\ g^{-1}$	change in lactate content in blood / mmol dm^{-3}
tailshaker muscle during rattling	+9.2	+0.148	+2.0
mammalian muscle during contraction	+5.0	+0.180	+5.0

Adapted from Kemper et al., PNAS, 2001

(c) Using the data from Table 3.1, explain why the tailshaker muscle is able to rely on anaerobic respiration for a longer time during rattling as compared to the mammalian muscle. [2]

Over the years, climate change has resulted in a rise in mean global temperatures. In Southwestern Europe, scientists collected data on the long-term population trends of a snake species and the rise in temperature in the region from 1980 to 2020, as shown in Fig. 3.2.

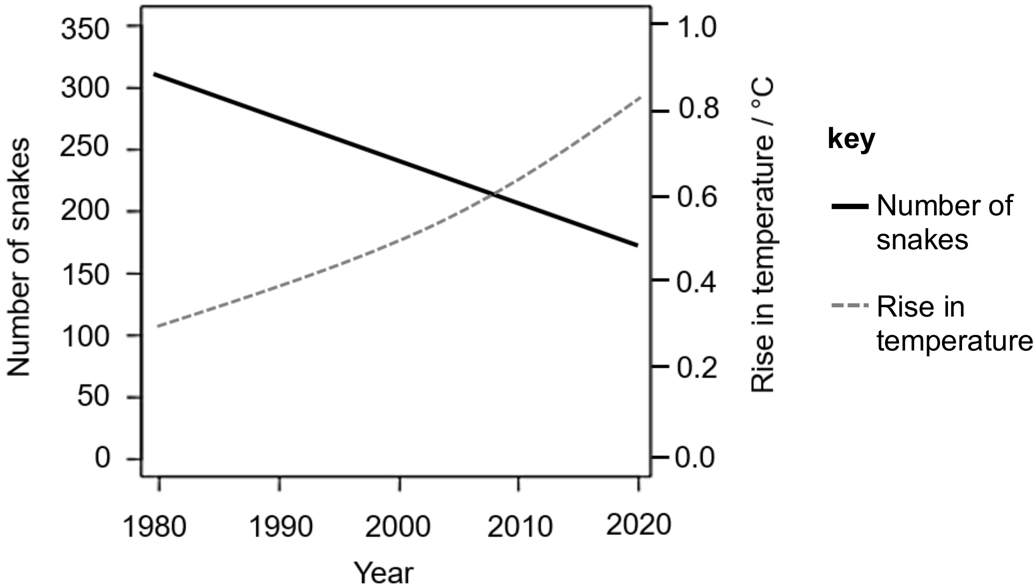

Fig. 3.2

(d) **(i)** With reference to Fig. 3.2, describe the impact of the rise in temperatures in Southwestern Europe on the number of snakes. [1]

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(ii) Suggest reasons for the change in number of snakes from 1980 to 2020. [2]

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[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

4 **(a)** Outline the functions of proteins involved in DNA replication and DNA condensation to form a chromosome in eukaryotes. [15]

(b) Describe the structure of viruses and explain the significance of viruses in evolution. [10]

[Total: 25]

5 **(a)** Outline the functions of proteins involved in processes which lower high blood glucose concentration. [15]

(b) Describe how the structure of antibody is related to its function and explain the significance of cell signaling in immune response. [10]

[Total: 25]

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